

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 9

SMALLEST AMONG TWO NUMBERS

AIM: To write and execute an 8085 assembly language program to find smallest among two numbers and store the result in memory.

ALGORITHM

1. Load the first number into register A.
2. Load the second number into register B.
3. Compare A with B.
4. If A is smaller, store A as the smallest.
5. Otherwise, move B to A.
6. Store the smallest number in memory.
7. Stop the program

CODE:

```
MVI A,25H  
MVI B,35H  
CMP B  
JC STORE_A  
MOV A,B  
STORE_A: STA 8050H  
HLT
```

OUTPUT:

FileResetAssemblerDebugHelp

Registers

A	03
BC	0A 00
DE	00 00
HL	00 00
PSW	00 00
PC	42 10
SP	FF FF
Int-Reg	00

Flag

S	1
Z	0
AC	0
P	1
C	1

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

2500

-

+

0A

Update Memory

Load me at

1

LDA 2500

2

MOV B, A

3

4

LDA 2501

5

CMP B

6

JG STORE

7

8

MOV A, B

9

10

STORE: STA 2502

11

HIT

DataStackKeyPadMemoryI/O Ports

Start2500

OK

Address (Hex)	Address	Data
09C4	2500	10
09C5	2501	3
09C6	2502	3
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No

Assembler Message

0

Program assembled successfully

Simulator: Idle

Search

ENG

IN

11:41

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